

RADHA LANGOJU

Data Scientist | LLM & Prompt Engineering | RAG & Agentic Systems
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PROFESSIONAL SUMMARY

Data Scientist who builds, optimizes, and deploys LLM-powered applications — spanning prompt engineering, retrieval-augmented generation (RAG), and agentic systems with OpenAI GPT, Hugging Face, and Python. This work builds on experience as an infrastructure and analytical engineer in public works, giving a rare grounding in how traditional industry operates. That combination, reinforced by published research, invited talks, and a track record of explaining complex systems to non-technical stakeholders, translates advanced AI into tools that work in real operational settings.

TECHNICAL SKILLS

LLM & GenAI: OpenAI GPT API, Hugging Face (Spaces, SmolAgents, Transformers), Lang Chain, Prompt Engineering & Optimization, RAG, Agentic / Tool-Use systems, Input guardrails

Languages & Frameworks: Python, SQL; PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

ML & Methods: NLP, Model Evaluation & Benchmarking, Forecasting, Physics-Informed Neural networks (PINNs), Computer Vision (YOLO)

Tools & Deployment: FastAPI, Git, Hugging Face, Deepgram (streaming STT), Tableau, Looker Studio

PROFESSIONAL EXPERIENCE

Data Scientist (Consultant) — Statistics Without Borders & nonprofit partners (Remote) Feb 2024 – Present
Statistics Without Borders · Good Data Institute · DataKind · UDO Foundation · University of Lagos

- Established reproducible pipelines, data-quality standards, and Tableau / Looker Studio dashboards for international data teams; led data engineering and analytics workshops for 90+ graduate students.
- Project & Client Manager**, manage the research partnership with the University of Lagos Dept. of Statistics — curating content and selecting speakers for a monthly international webinar series, driving knowledge-sharing across a global academic audience.

AI Data Scientist (Volunteer) — Facile AI (Remote) Jun 2025 – Jan 2026
Kammi, a multimodal medical voice assistant for orthopedic intake. [Live demo \(HF Spaces\)](#)

- Built an end-to-end voice assistant answering patient queries on knee-replacement surgery from a clinician-approved knowledge base, making clinical information accessible through natural conversation.
- Designed and deployed an end-to-end pipeline integrating streaming speech recognition, retrieval-augmented generation, and text-to-speech, reaching 95%+ transcription accuracy and 98% chunk-retrieval recall over the clinical knowledge base.
- Reduced dialogue latency 30% via streaming token-level LLM outputs directly to TTS, improving real-time experience.
- Implemented context-aware conversation memory for coherent multi-turn interactions.

Assistant Executive Engineer July 2017 – Aug 2023
Quality Control Division, Public Health & Municipal Engineering Department

- Directed data-driven optimization of multi-city water-supply infrastructure and modernization of 200 state schools, improving enrollment rates by 15% — deep exposure to **traditional industry operations and challenges**.
- Presented quantitative analysis on multimillion-dollar capital projects to senior government stakeholders, adapting to changing project requirements in a collaborative team environment.

RESEARCH & INVITED TALKS

- Conference Paper** – “Inverse Physics-Informed Neural Network Framework for Planetary Gear Parameter Identification using Experimental Data and Embedded Load Distribution Physics,” ASME IDETC-CIE Conference 2026. Designed the Neural Network architecture and dual residual training to infer latent system parameters by coupling experimental measurements with a non-differentiable physics solver. ([Link to Abstract](#))
- Invited Speaker, University of Lagos, Dept. of Statistics (2026) and American Statistical Association, Albany Chapter (2025)** — delivered talks on the replication crisis and statistical reproducibility to international student and faculty audiences, articulating complex methodology to non-technical listeners.

SELECTED PROJECTS

- Banking Natural-Language Query Agent:** Agentic system translating natural-language questions into validated SQL using Hugging Face SmolAgents with input guardrails — 89% query accuracy on the evaluation set.
- YouTube Transcript RAG Q&A:** Local-first retrieval-augmented application for summarization and grounded question-answering over video content with on-device inference.
- Dental Radiograph Object Detection:** YOLO computer-vision model detecting missing teeth and mandibular canals across 200+ radiographs — 90.5% precision, 85.7% recall.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering, Civil Engineering — Andhra University College of Engineering, India (2012–2016)

Selected Certifications: IBM RAG and Agentic AI Specialization (2026); Google Advanced Data Analytics (2025); TensorFlow Developer Certificate (2024); Machine Learning Specialization, Stanford / DeepLearning.AI.; Deep learning Specialization, IIT Madras (2024)